

Example

Use the data linearization method to fit the curve of the form $f(x) = ax^b$ using Least Square Approximation method.

x_k	1	2	3	4
y_k	0.6	1.9	4.3	7.6

$$y = ax^b, \ln(y) = \ln(ax^b) = \ln(a) + b \ln(x)$$

$$Y = \ln(y) \quad X = \ln(x), \quad A = \ln(a)$$

k	x_k	y_k	X_k	Y_k	$X_k Y_k$	X_k^2
1	1	0.6	0	-0.510	0	0
2	2	1.9	0.693	0.641	0.444	0.480
3	3	4.3	1.098	1.458	1.600	1.205
4	4	7.6	1.386	2.028	2.810	1.920
Sum			3.177	3.617	4.854	3.605

$$E = \sum (Y_i - A - bX_i)^2$$

$$\sum Y_i = A \sum 1 + b \sum X_i$$

$$\sum Y_i X_i = A \sum X_i + b \sum X_i^2$$

$$3.617 = 4A + 3.177b$$

$$4.854 = 3.177A + 3.605b$$

$$A = \frac{\frac{3.617}{4} - \frac{3.177}{3.177} \cdot \frac{3.605}{3.177}}{\frac{3.177}{4} - \frac{3.605}{3.177}} = \frac{-2.381}{4.326} = -0.550 \quad A = \ln(a) \quad a = \exp(A) \quad a = \exp(-0.550) = 0.576$$

$$b = \frac{\frac{3.177}{4} - \frac{4.854}{3.177}}{\frac{3.177}{4} - \frac{3.605}{3.177}} = \frac{7.924}{4.326} = 1.831$$

$$f(x) = ax^b = 0.566x^{1.831}$$

